AI Playbook and Toolkit for Public Health Departments

Natural Language Processing for Public Health

ANL 200

Natural language processing is a branch of AI that works with human language. Public health agencies often hold important information in free-text sources, including clinical notes, case narratives, call center logs, inspection reports, complaints, death certificate text, public comments, and community feedback. This module teaches learners how NLP can support public health work and how to evaluate its risks.

Level	applied/foundational
Primary track	Analytics and Modeling Track
Appears in tracks	technical-architecture

Learning Objectives

- Define NLP in plain language and distinguish common NLP tasks.
- Identify public health workflows where NLP may be useful.
- Assess risks related to accuracy, bias, privacy, hallucination, language, and missing context.
- Define source verification and human review requirements for NLP outputs.
- Produce an NLP evaluation plan for one public health workflow.

Module Overview

Natural language processing is a branch of AI that works with human language. Public health agencies often hold important information in free-text sources, including clinical notes, case narratives, call center logs, inspection reports, complaints, death certificate text, public comments, and community feedback. This module teaches learners how NLP can support public health work and how to evaluate its risks.

Why This Topic Matters for Public Health AI

NLP can reduce burden by helping staff find, organize, classify, extract, translate, or summarize information from large volumes of text. It can support case investigation, environmental health complaint triage, syndromic surveillance, public inquiry response, community engagement analysis, and policy review. However, text is often ambiguous, incomplete, sensitive, and context-dependent.

NLP outputs should not be treated as facts without review. A tool may omit important details, misread abbreviations, misunderstand negation, translate poorly, or generate a summary that sounds correct but is not supported by the source. Public health text may also include legal, clinical, cultural, or community context that a model does not understand.

For technical staff, NLP evaluation requires more than checking whether outputs sound fluent. The agency should define the task, intended use, prohibited use, source material, review process, performance measures, subgroup considerations, privacy safeguards, and escalation pathway.

Core Concepts

Natural language processing. NLP refers to AI methods that process, analyze, classify, summarize, extract, translate, or generate human language. It can work with structured prompts, free-text notes, documents, emails, calls, complaints, or reports.

Information extraction. Information extraction pulls specific facts from text, such as symptoms, dates, locations, exposures, risk factors, facility names, or requested services. Extraction should be validated against source text.

Text classification. Text classification assigns text to categories, such as complaint type, urgency, topic, likely condition, or program area. Classification requires labeled examples, clear categories, and evaluation.

Summarization. Summarization creates a shorter version of longer text. Public health summarization should preserve uncertainty, cite sources when possible, and avoid adding unsupported claims.

Hallucination. A hallucination is an AI-generated statement that is false, unsupported, or not present in the source material. Hallucination risk is especially important when summaries influence public health decisions or communications.

Public Health Example

A contact center may receive hundreds of public inquiries during an outbreak. NLP could classify messages by topic, identify urgent issues, and draft response summaries for staff review. This can improve timeliness, but messages may contain sensitive information, emotional language, misinformation, or ambiguous requests. Staff need review procedures and escalation criteria.

An environmental health program may use NLP to classify complaints about food safety, housing, water, pests, or nuisance conditions. The tool may help route complaints, but errors can affect response time and enforcement. The program should validate classification accuracy and ensure that original complaint text remains available to reviewers.

Technical Deep Dive

NLP implementation begins with task definition. Extraction, classification, summarization, translation, topic modeling, and question answering are different tasks with different evaluation methods. A tool that performs well at summarization may perform poorly at extracting dates or classifying urgency. Staff should define the task narrowly before selecting or testing a tool.

Evaluation should use representative source text. Test examples should include common language, uncommon language, abbreviations, misspellings, multilingual text, ambiguous statements, negation, sensitive information, and edge cases. Performance should be reviewed by people who understand the public health workflow, not only by technical staff.

Privacy and source traceability are central. NLP may process sensitive case notes, complaints, or community feedback. The agency must define whether the data can be used, where processing occurs, how outputs are stored, who can access them, and how source text is preserved. For consequential uses, users should be able to trace extracted facts or summaries back to the original text.

Risks, Failure Modes, and Guardrails

Unsupported summaries. NLP tools may omit, distort, or invent information. Guardrails include source review, citations, factuality checks, and human approval.

Language and accessibility gaps. Models may perform differently across languages, dialects, literacy levels, or disability-related communication needs. Guardrails include multilingual evaluation, accessibility review, and community context.

Privacy exposure. Free text may contain sensitive or unexpected personal information. Guardrails include approved environments, de-identification when appropriate, access controls, and retention rules.

Misclassification. Incorrect labels can affect routing or priority. Guardrails include validation, confidence thresholds, review queues, and monitoring of overrides.

Application to AI-Supported Workflows

Generative AI is often used for summarization, drafting, and question answering, but outputs must be checked against source material. Predictive analytics may use NLP-derived features, so extraction quality affects model performance. RAG systems use NLP to retrieve and summarize documents, which requires document governance and source citation. Agentic AI using NLP to trigger actions should require clear thresholds and human approval for consequential steps.

Reflection Questions

Which public health workflows involve large volumes of free text?

What task would NLP support: extraction, classification, summarization, translation, or topic detection?

What errors would be most harmful?

What source material must be retained for human review?

How will staff evaluate accuracy across languages, populations, and program areas?

Practical Exercise

Create an NLP evaluation plan for one public health workflow. Include the text source, NLP task, intended use, prohibited use, required human review, accuracy measures, bias or subgroup review, privacy safeguards, source citation or traceability expectations, and escalation process.

Expected Artifact or Evidence

NLP use case description

NLP evaluation plan

Human review checklist

Source verification and privacy safeguards

Expected Assignment

- NLP use case description
- NLP evaluation plan
- Human review checklist
- Source verification and privacy safeguards

Knowledge Check

- 1. What is NLP?
- 2. Which task is an example of information extraction?
- 3. Why is source verification important for NLP outputs?
- 4. Which NLP use requires strong safeguards?
- 5. What is strong completion evidence for this module?

References and Resources

- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST AI RMF Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- WHO Guidance on Large Multi-Modal Models: https://www.who.int/publications/i/item/9789240084759
- CDC Public Health Data Strategy: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- Agency privacy, records, data governance, and program policies